

n8n Workshop

Hands-On Build Guide

AI-Powered Support Ticket Triage System

9 Nodes

AI Agent

OpenRouter

Slack +
Gmail

Google
Sheets

Live Triage

Workflow Overview

This workflow receives support tickets via a webhook, uses an AI Agent to classify urgency and draft a reply, routes HIGH urgency tickets to Slack while LOW urgency ones get an auto-email, and logs every ticket to Google Sheets.

Node sequence

Step	Node name
01	Webhook – receives POST /support-ticket
02	Edit Fields (Set) – extract & rename fields
03	AI Agent – classify + draft reply
03a	OpenRouter Chat Model – LLM sub-node
03b	Structured Output Parser – enforce JSON shape
04	IF node – route by urgency
05a	Send a message (Slack) – HIGH branch
05b	Send a message1 (Gmail) – LOW branch
06	Append row in sheet (Google Sheets) – log all

Node 01 : Webhook — Trigger

Listens for incoming POST requests from the support form. This is the entry point of the entire workflow.

Configuration

Parameter	Value
Node type	webhook
HTTP method	POST
Path	/support-ticket
Respond	Immediately
Response code	200
Response data	First entry JSON

Expected sample incoming JSON body

```
{
  "name": "Priya Sharma",
  "email": "priya.sharma@gmail.com",
  "subject": "Charged twice for subscription",
  "message": "Hi, I was charged twice this month..."
}
```

How to test

1. Click Listen for test event in the Webhook node
2. Open Postman → POST → <http://localhost:5678/webhook/support-ticket>
3. Set header: Content-Type: application/json
4. Paste the JSON above into the Body tab → Send
5. n8n should show the incoming data in green

Node 02 : Set node Extract & rename incoming fields cleanly

Extracts and renames the raw webhook body fields into clean, readable variable names used by all downstream nodes.

Configuration

Parameter	Value
Node type	Edit Fields
Mode	Manually map fields
Keep others	Off

Field mappings (set all 5)

Field name	Expression
customerName	={{ \$json.body.name }}
customerEmail	={{ \$json.body.email }}
ticketSubject	={{ \$json.body.subject }}
ticketMessage	={{ \$json.body.message }}
receivedAt	={{ \$now.toISO() }}

Node 03 : AI Agent node Classify urgency + write draft reply

The core AI node. Reads the support message, classifies urgency and category, and writes a professional email reply. Uses three connected sub-nodes: a language model, a memory buffer, and a structured output parser.

Agent node configuration

Parameter	Value
Node type	AI Agent node
Agent type	Conversational Agent
Prompt type	Define (not from input)
User message	{{ \$json.ticketMessage }}
Has output parser	true (enable toggle)

System prompt — paste this exactly

```
You are a customer support triage assistant.

Analyze the customer message and return ONLY valid JSON
with exactly these 4 keys:

1. "urgency": "HIGH" or "LOW"
  - HIGH = payment issues, account locked, data loss, service down
  - LOW = general questions, feature requests, feedback

2. "category": one of "billing", "technical", "account", "general"

3. "reply": A professional, empathetic email reply to the customer.
  Start with "Hi [customer name]," – use the name from context.

4. "summary": A 1-sentence internal summary for the support team.

Return ONLY the JSON object. No markdown, no extra text.
```

Sub-node A: OpenRouter Chat Model

Parameter	Value
Node type	OpenRouter Chat Model
Model	nvidia/nemotron-nano-9b-v2:free

Credential	OpenRouter account (OpenRouter API key)
Temperature	0.3 (recommended for consistent JSON)
Max tokens	1000

Sub-node B: Structured Output Parser

Parameter	Value
Node type	Output Parser Structured
Schema type	From JSON example
Retry on fail	Yes (auto-fix enabled)

Output parser JSON schema example

```
{
  "urgency": "HIGH",
  "category": "billing",
  "reply": "Hi Priya, thank you for reaching out...",
  "summary": "Customer charged twice, refund required."
}
```

How to access AI output in downstream nodes

Field needed	Expression to use
Urgency level	={{ \$json.output.urgency }}
Category	={{ \$json.output.category }}
AI reply text	={{ \$json.output.reply }}
Internal summary	={{ \$json.output.summary }}

Node 04 : IF node Route by urgency level

Reads the urgency field from the AI Agent output and splits execution into two branches: TRUE for HIGH urgency (Slack alert) and FALSE for LOW urgency (Gmail reply).

Configuration

Parameter	Value
Node type	IF
Value 1	={{ \$json.output.urgency }}
Operator	String → equals
Value 2	HIGH
Case sensitive	Yes
Combinator	AND

Branch behaviour

Branch	Condition
TRUE (output 0)	urgency = HIGH → goes to Slack node
FALSE (output 1)	urgency = LOW → goes to Gmail node

Node 05a : Slack node (HIGH branch)

Fires only on the TRUE branch. Sends a formatted alert message to the #support-urgent Slack channel so the team can respond immediately.

Configuration

Parameter	Value
Node type	Slack -- send a message
Resource	Message
Operation	Send
Credential	Slack OAuth2 API (connect workspace)
Select	Channel
Channel ID	pick #support-urgent from list

Message text — paste this exactly

```
*HIGH urgency ticket*

*From:* {{ $json.customerName }} ({{ $json.customerEmail }})
*Category:* {{ $json.output.category }}
*Summary:* {{ $json.output.summary }}
*Received:* {{ $json.receivedAt }}
```


Node 05b : Gmail node (LOW branch)

Fires only on the FALSE branch. Sends the AI-drafted reply directly to the customer's email address, using the subject from their original ticket.

Configuration

Parameter	Value
Node type	Gmail -- send a message
Resource	Message
Operation	Send
Credential	Gmail OAuth2 API (connect Google account)
Send To	{{ \$('Edit Fields').item.json.customerEmail }}
Subject	{{ \$('Edit Fields').item.json.ticketSubject }}
Email type	Text (or HTML)
Message	{{ \$json.output.reply }}

Why reference Edit Fields here?

After the IF node, \$json refers to the IF node's output — which only has output.* fields. To get customerEmail and ticketSubject, we reference the earlier Edit Fields node directly using \$('Edit Fields').item.json.*

Node 06 : Google Sheets node Log every ticket for reporting

Receives output from BOTH Slack and Gmail branches (no Merge node needed — both connect to this node's input). Appends one row per ticket to the tracking spreadsheet.

Configuration

Parameter	Value
Node type	Google Sheets -- append row
Resource	Sheet within document
Operation	Append or update row
Credential	Google Sheets OAuth2 API
Document	Select your spreadsheet from Google Drive
Sheet	Sheet3 (or the sheet with headers in row 1)
Mapping mode	Define below (manual)

Column mappings — set all 8 columns

Column header	Expression
Timestamp	={{ \$now }}
Name	={{ \$('Edit Fields').item.json.customerName }}
Email	={{ \$('Edit Fields').item.json.customerEmail }}
Subject	={{ \$('Edit Fields').item.json.ticketSubject }}
Category	={{ \$('If').item.json.output.category }}
Urgency	={{ \$('If').item.json.output.urgency }}
Summary	={{ \$('If').item.json.output.summary }}
AI Reply	={{ \$('If').item.json.output.reply }}

Test Payloads

Use these JSON payloads in Postman to test all branches. Set Method=POST, URL=http://localhost:5678/webhook/support-ticket, Header: Content-Type=application/json.

HIGH urgency — Billing (expect: Slack alert)

```
{
  "name":    "Priya Sharma",
  "email":   "priya.sharma@gmail.com",
  "subject": "Charged twice for my subscription",
  "message": "Hi, I was charged Rs.4,999 twice this month on
              1st April and again on 3rd April. I have not received
              any refund. Please resolve immediately."
}
```

HIGH urgency — Account locked (expect: Slack alert)

```
{
  "name":    "Rahul Deshmukh",
  "email":   "rahul.d@outlook.com",
  "subject": "Cannot login – account locked",
  "message": "I have been locked out of my account since yesterday.
              I tried resetting my password three times. I have an
              important presentation tomorrow, please unlock immediately."
}
```

LOW urgency — General query (expect: Gmail auto-reply)

```
{
  "name":    "Sneha Kulkarni",
  "email":   "sneha.k@yahoo.com",
  "subject": "How to export my data?",
  "message": "Hello, I wanted to know if there is a way to export
              all my data as a CSV or Excel file. I have been using
              your service for 6 months. Thank you."
}
```

LOW urgency — Technical bug (expect: Gmail auto-reply)

```
{
  "name":    "Amit Joshi",
  "email":   "amitjoshi92@gmail.com",
  "subject": "Dashboard not loading on mobile",
  "message": "Hi, the dashboard does not load properly on Chrome
              Android. The sidebar overlaps buttons. Works fine on
              desktop. Running Chrome 124 on Android 13."
}
```

LOW urgency — Positive feedback (expect: Gmail auto-reply)

```
{
  "name":    "Meena Patil",
  "email":   "meenapatil@hotmail.com",
  "subject": "Love the new update!",
  "message": "The new dark mode and performance improvements are
              fantastic. Suggestion: can you add keyboard shortcuts
              to switch between projects quickly?"
}
```

Quick Reference

All node expressions at a glance

What you need	Expression
Customer name (from Set node)	<code>={{ \$('Edit Fields').item.json.customerName }}</code>
Customer email (from Set node)	<code>={{ \$('Edit Fields').item.json.customerEmail }}</code>
Ticket subject (from Set node)	<code>={{ \$('Edit Fields').item.json.ticketSubject }}</code>
Raw ticket message	<code>={{ \$('Edit Fields').item.json.ticketMessage }}</code>
AI urgency (HIGH/LOW)	<code>={{ \$json.output.urgency }}</code>
AI category	<code>={{ \$json.output.category }}</code>
AI-written reply	<code>={{ \$json.output.reply }}</code>
AI internal summary	<code>={{ \$json.output.summary }}</code>
Urgency from IF node (in Sheets)	<code>={{ \$('If').item.json.output.urgency }}</code>
Category from IF node (in Sheets)	<code>={{ \$('If').item.json.output.category }}</code>

Common errors and fixes

Error	Fix
Output parser fails / empty response	Check system prompt ends with 'Return ONLY the JSON object'
Gmail node: expression undefined	Use <code>\$('Edit Fields').item.json.customerEmail</code> not <code>\$json.customerEmail</code>
Sheets: columns not mapping	Ensure Row 1 headers match exactly: Timestamp, Name, Email...
Slack: channel not found	Use channel ID (e.g. C0AV307FZLY) instead of #channel-name
IF node: both branches not triggering	Test HIGH and LOW payloads separately – only one fires per run
AI returns markdown instead of JSON	Add 'No markdown, no extra text' at end of system prompt

Credentials checklist

Credential	Node
OpenRouter API key	OpenRouter Chat Model
Slack OAuth2 API	Send a message (Slack)
Gmail OAuth2 API	Send a message1 (Gmail)
Google Sheets OAuth2 API	Append row in sheet